

2-sided Inverse

$$AA^T = I = A^T A$$

$$\text{rank} = m = n$$

Left Inverse

$$\text{Rank} = n \rightarrow \text{Full col Rank}$$

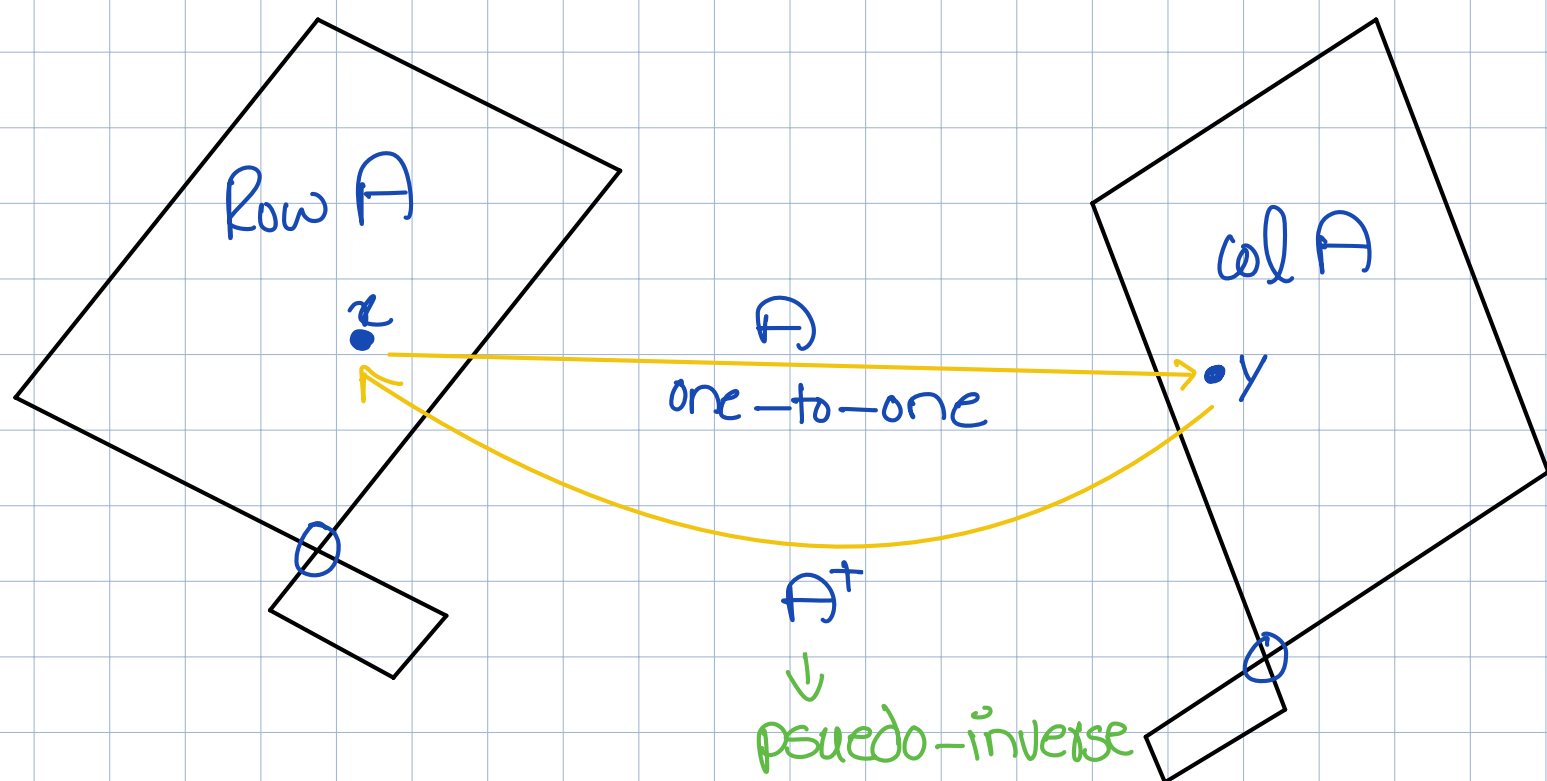
$$\text{Null } A = \{0\}$$

Independent Columns

0 or 1 sols to $Ax=b$

$$\underbrace{\left[(A^T A)^{-1} A^T \right]}_{A_{\text{left}}^{-1}} A = I_{n \times n}$$

Pseudo-inverse:



If $x \neq y \in \text{Row } A$ then $Ax \neq Ay$

Suppose $Ax = Ay$

$$A(x-y) = 0$$

$\hookrightarrow$ Null space

$\hookrightarrow$ As $x, y \in \text{Row } A$ so $x-y \in \text{Row } A$

Right Inverse

$$\text{Rank} = m \rightarrow \text{Full row Rank}$$

Independent Rows

$$\text{Null } A^T = \{0\}$$

$Ax=b$ has always a sol

$$A \underbrace{\left[A^T (A A^T)^{-1} \right]}_{A_{\text{right}}^{-1}} = I$$

Which is only when $x=y=0$

Find A^+ :

1. $A = U \Sigma V$

$m \times n$

Rank = r

$$\Sigma^+ = \begin{bmatrix} 1/\sigma_1 & & & 0 \\ & 1/\sigma_2 & & \\ & & \ddots & \\ 0 & & & 1/\sigma_r & \\ & & & & 0 \end{bmatrix}$$

$$\Sigma \Sigma^+ = \begin{bmatrix} 1 & & & 0 \\ & 1 & & \\ & & \ddots & \\ 0 & & & 1 & \\ & & & & 0 \end{bmatrix}$$

$m \times m$

$$\Sigma^+ \Sigma = \begin{bmatrix} 1 & & & 0 \\ & 1 & & \\ & & \ddots & \\ 0 & & & 1 & \\ & & & & 0 \end{bmatrix}$$

$n \times n$

Projection of Row & col &
Removes Null space